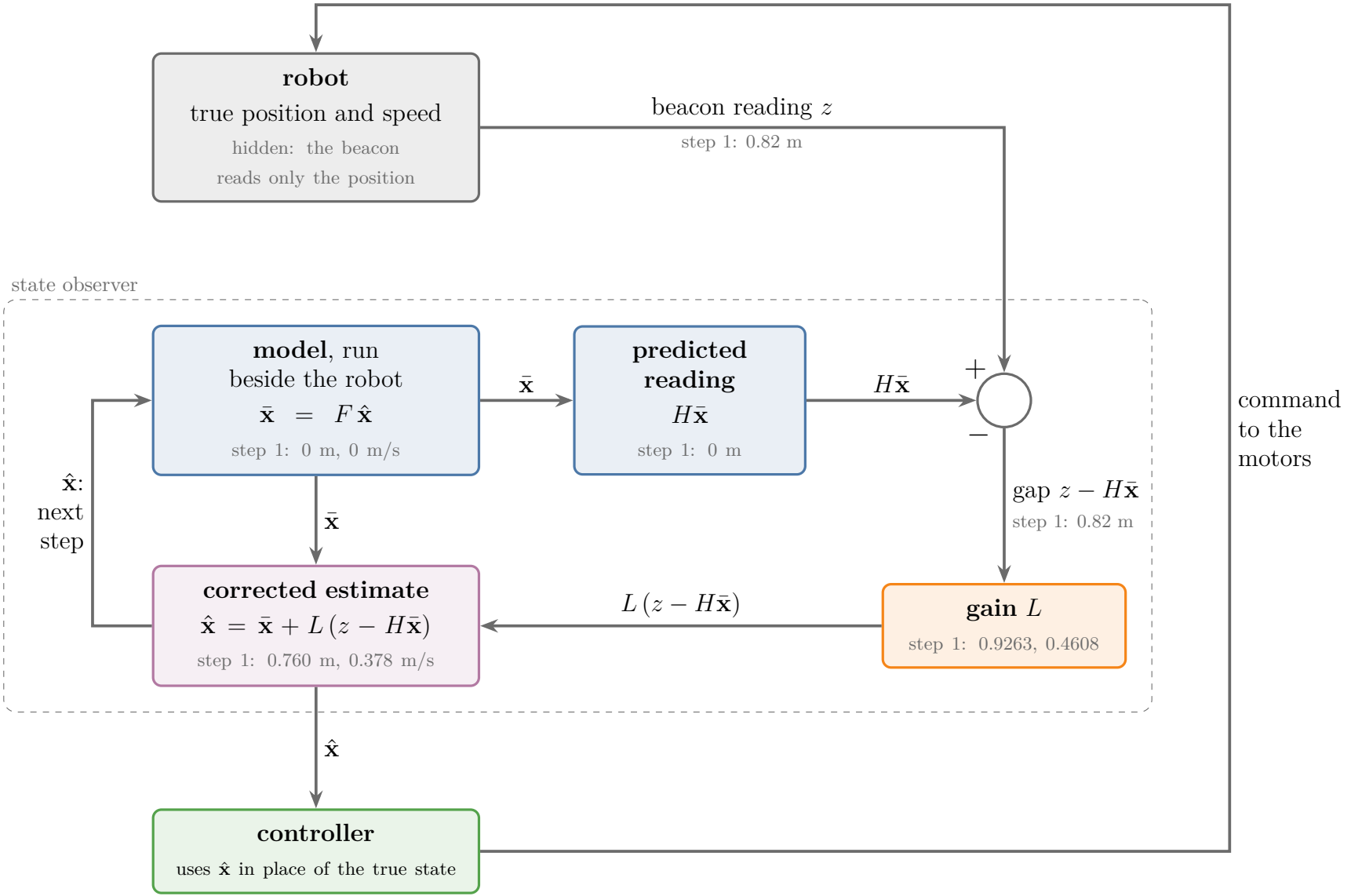

robot
true position and speed
hidden: the beacon
reads only the position
beacon reading z
step 1: 0.82 m
state observer
model, run
beside the robot
$\bar{\mathbf{x}} = F\,\hat{\mathbf{x}}$
step 1: 0 m, 0 m/s
$\bar{\mathbf{x}}$
predicted
reading
$H\bar{\mathbf{x}}$
step 1: 0 m
$H\bar{\mathbf{x}}$
+
−
gap $z - H\bar{\mathbf{x}}$
step 1: 0.82 m
command
to the
motors
$\hat{\mathbf{x}}$:
next
step
$\bar{\mathbf{x}}$
corrected estimate
$\hat{\mathbf{x}} = \bar{\mathbf{x}} + L\,(z - H\bar{\mathbf{x}})$
step 1: 0.760 m, 0.378 m/s
$L\,(z - H\bar{\mathbf{x}})$
gain L
step 1: 0.9263, 0.4608
$\hat{\mathbf{x}}$
controller
uses $\hat{\mathbf{x}}$ in place of the true state